responsibility for actions taken by an autonomous drone, the operator, the manufacturer, or the AI system itself, poses serious moral and legal dilemmas.

5.8 Case Study: Operational Deployment and Impact of Fourth-generation Drones

The emergence of fourth-generation drones marks a transformative phase in global defense, surveillance, and civil operations defined by autonomous decision-making, AI-based swarm intelligence, stealth technology, and realtime data integration. These case studies examine the operational deployment and impact of these advanced UAV systems through key examples across military, humanitarian, and scientific domains, highlighting how technological integration and strategic utilization have reshaped modern aerial operations.

5.8.1 Case study 1: Military operations: The XQ-58A

Valkyrie and Skyborg Integration (2020–present)

In one of the most prominent deployments, the US Air Force tested the XQ-58A Valkyrie under the Skyborg Program to establish AI-enabled manned–unmanned teaming (MUM-T). The Valkyrie operated alongside F-35 and F-15EX fighters, executing autonomous reconnaissance, electronic warfare, and simulated combat support missions.

The drone’s AI autonomy core allowed it to make tactical decisions such as threat avoidance and target prioritization without direct human input, demonstrating the feasibility of AI-led battlefield collaboration. Operational results revealed a 35% increase in mission survivability and a 50% reduction in pilot workload, proving the strategic effectiveness of autonomous combat support drones.